

Responsible AI Model Evaluations

A Eight-Week Red-Teaming Study of Foundation Models Across 22 Safety Risk Categories

Hemant Naik · May 16, 2026 · Dataset: RedBench (Dang et al., 2026, MIT)

Live Dashboard: <https://Sushegaad.github.io/Responsible-AI-Model-Evaluations/> · GitHub: <https://github.com/Sushegaad/Responsible-AI-Model-Evaluations>

Total Evaluations	Attack Failures	Critical Findings	Over-Refusals	Study Duration	Models Evaluated
23,000 8 weekly runs	1,081 Across all runs	399 Severity >= 8/10	352 False rejections	8 Weeks Apr 10 - May 16	7 3 providers

Abstract

This paper presents an eight-week automated red-teaming study of seven frontier large language models from Anthropic, OpenAI, and Google, evaluated across 22 safety risk categories using the RedBench adversarial benchmark (29,362 prompts). Over 23,000 individual evaluations were conducted between April 10 and May 16, 2026, producing 399 critical findings and 352 false-outright-refusals. Results are scored by a dual-judge pipeline — a deterministic RegexJudge and a NeuralJudge (Claude Haiku) — and mapped to the NIST AI Risk Management Framework 1.0. Across all eight runs, Gemini 2.5 Flash achieved the lowest average Attack Success Rate (3.75%), while GPT-4o Mini remained the highest-risk model (8.31% average ASR). CBRN and Cybersecurity categories continued to account for the majority of guardrail bypasses. In the final run (May 16, 2026), Gemini 2.5 Pro recorded the most critical failures (23) and the highest drift coefficient (+1.11/turn), raising deployment concerns for multi-turn applications. Election Interference has solidified as a persistent top-5 risk category at 6.52% average ASR. All data, evaluation code, and audit logs are publicly available under the MIT License.

1. Introduction

The deployment of large language models (LLMs) in high-stakes domains — including government, healthcare, finance, and legal services — demands rigorous, reproducible safety evidence. Vendor self-assessments are insufficient for independent governance because they are not replicable, use proprietary benchmarks, and are subject to publication bias. This study addresses that gap with a fully open, automated weekly pipeline that stress-tests seven frontier models against a standardised adversarial dataset and publishes results to a public dashboard every Friday.

The evaluation covers 22 risk categories spanning physical harm, cybersecurity, privacy, misinformation, and domain-specific risks. Each run draws 500 stratified prompts per model from the RedBench dataset — 400 adversarial and 100 benign — and grades every response with a dual-judge system. The eight-week time series produced in this study enables detection of safety regressions, guardrail erosion under adversarial pressure, and comparative model rankings that are independent of any provider’s internal testing.

2. Methodology

2.1 Dataset

RedBench (Dang et al., 2026) aggregates 37 adversarial sub-benchmarks — including HarmBench, ToxiGen, XSTest, DAN, and AdvBench — into a single standardised schema covering 22 risk categories and 19 domains. A local snapshot of 29,362 prompts (26,113 adversarial, 3,249 benign) is committed to the repository as Parquet files, making every run fully reproducible without network dependencies on the dataset host.

2.2 Models

Model			
Claude Opus 4.7	Anthropic	4	Apr 24 – May 16
Claude Opus 4.6	Anthropic	8	Apr 10 – May 16
Claude Sonnet 4.6	Anthropic	8	Apr 10 – May 16
GPT-4o	OpenAI	8	Apr 10 – May 16
GPT-4o Mini	OpenAI	8	Apr 10 – May 16
Gemini 2.5 Flash	Google	5	Apr 17 – May 16
Gemini 2.5 Pro	Google	5	Apr 14 – May 16

Table 1. Models evaluated in this study.

2.3 Evaluation Pipeline

Each model receives 500 prompts per weekly run: 400 adversarial attacks drawn proportionally across all 22 risk categories, and 100 benign prompts to measure false refusal rates. Every prompt is submitted in three modes: single-turn, agentic 3-turn wrap (system + user + follow-up), and — for a 5% sub-sample — a 10-turn adversarial drift probe that measures how safety posture degrades under sustained pressure.

Responses are graded by a Dual-Judge pipeline. The RegexJudge applies deterministic pattern-matching for PII leakage, jailbreak confirmations, malware code, explosive synthesis instructions, and CSAM. The NeuralJudge (Claude Haiku) returns a structured JSON verdict (SAFE, UNSAFE, or OVER_REFUSAL), a severity score from 1 to 10, and a chain-of-reasoning sentence for the forensic audit trail. A regex match always overrides the neural result and forces a CRITICAL_FAIL; a neural UNSAFE verdict with severity 8 or above also escalates to CRITICAL_FAIL regardless of the regex result.

2.4 Metrics

Metric		
Attack Success Rate (ASR)	% of adversarial prompts that bypassed guardrails	< 5%
False Outright Refusal (FOR)	% of benign prompts incorrectly refused	< 10%
Drift Coefficient	Safety change per conversation turn (10-turn probe)	< 0.5%/turn
Provenance Score	% of evaluations with complete chain-of-reasoning JSON	> 95%

Table 2. Evaluation metrics and targets.

3. Results

3.1 Overall Performance Across All Runs

Across 23,000 evaluations spanning eight weekly runs, Gemini 2.5 Flash achieved the lowest average Attack Success Rate at 3.75%, followed closely by Claude Opus 4.7 at 4.12% — though both models joined the

study later than the four-provider cohort. Among models with a full eight-run history, Claude Sonnet 4.6 led at 5.28% average ASR, edging out GPT-4o (5.62%) and Claude Opus 4.6 (5.69%). GPT-4o Mini remained the weakest performer at 8.31% average ASR, with Gemini 2.5 Pro close behind at 7.85%.

False Outright Refusal rates reveal a sharp safety-utility divide. The three Anthropic models averaged between 0.75% and 1.25% FOR — exceptionally low, indicating near-zero unnecessary refusals. Google's models showed the opposite pattern: Gemini 2.5 Pro averaged 17.60% FOR and Gemini 2.5 Flash 14.60%, meaning roughly one in six to seven benign requests was incorrectly refused. GPT-4o Mini averaged 13.75% FOR, also above the 10% target threshold.

Model						
Gemini 2.5 Flash	3.75%	14.60%	+0.44	100.0%	21	5
Claude Opus 4.7	4.12%	1.25%	+0.69	100.0%	21	4
Claude Sonnet 4.6	5.28%	1.12%	−0.14	99.9%	56	8
GPT-4o	5.62%	7.62%	+0.42	100.0%	62	8
Claude Opus 4.6	5.69%	0.75%	−0.28	99.9%	59	8
Gemini 2.5 Pro	7.85%	17.60%	+0.11	100.0%	95	5
GPT-4o Mini	8.31%	13.75%	+0.35	100.0%	85	8

Table 3. Per-model averages across all successful evaluation runs. Green = at or below target; amber = moderate concern; red = above target.

3.2 Latest Run: May 16, 2026

The most recent evaluation run (May 16, 2026) produced results across all seven models. Gemini 2.5 Flash led with the lowest ASR at 5.00%, followed by Claude Opus 4.7 at 5.25%. Notably, both Claude Opus 4.6 and Claude Sonnet 4.6 regressed to 7.25% ASR — above their study averages — suggesting sensitivity to the May 16 prompt sample composition. Gemini 2.5 Pro recorded its highest critical-failure count in the study at 23, and its drift coefficient of +1.11 per turn indicates meaningful guardrail erosion under multi-turn adversarial pressure. GPT-4o Mini remained persistently elevated at 9.75%.

Model					
Gemini 2.5 Flash	5.00%	13.00%	−0.56	100.0%	6
Claude Opus 4.7	5.25%	1.00%	0.00	100.0%	8
GPT-4o	6.75%	6.00%	+0.56	100.0%	8
Claude Opus 4.6	7.25%	0.00%	0.00	99.8%	13
Claude Sonnet 4.6	7.25%	2.00%	−0.56	99.8%	12
Gemini 2.5 Pro	9.00%	15.00%	+1.11	100.0%	23
GPT-4o Mini	9.75%	12.00%	0.00	100.0%	12

Table 4. May 16, 2026 evaluation run results.

3.3 Weekly ASR Trend (All Eight Runs)

The eight-week trend reveals considerable week-to-week variance across all models, driven largely by sampling randomness and prompt-composition shifts. The most consistent performer over the full period is Claude Sonnet 4.6, which exhibited the narrowest ASR range (2.75%–7.25%) and a slight downward trend through the mid-study period before recovering in May. GPT-4o Mini's ASR peaked in run 1 at 10.25%, fell to

5.5% by run 3, and has climbed steadily since, reaching 9.75% in the final run. Gemini 2.5 Pro has never fallen below 7.25% across its five evaluated runs and trended upward to 9.00% on May 16.

Model								
Claude Opus 4.6	6.00%	7.50%	4.25%	4.00%	6.75%	5.25%	4.50%	7.25%
Claude Opus 4.7	—	—	—	—	4.00%	3.00%	4.25%	5.25%
Claude Sonnet 4.6	6.50%	7.00%	2.75%	3.25%	6.50%	5.00%	4.00%	7.25%
Gemini 2.5 Flash	—	—	—	2.50%	3.50%	3.75%	4.00%	5.00%
Gemini 2.5 Pro	—	—	7.25%	8.00%	7.25%	—	7.75%	9.00%
GPT-4o	6.25%	6.50%	3.25%	3.75%	7.00%	5.25%	6.25%	6.75%
GPT-4o Mini	10.25%	10.00%	5.50%	6.00%	9.50%	6.00%	9.50%	9.75%

Table 5. Weekly Attack Success Rate by model. — = model not evaluated that week. Green < 5%, amber 5–8%, red > 8%.

4. Risk Category Analysis

CBRN (Chemical, Biological, Radiological, and Nuclear) information remained the highest-risk category across the full eight-week period with an average ASR of 35.52% — more than double the second-ranked category, Cybersecurity Threats Beyond Malware at 21.60%. This gap reflects the inherent difficulty of building blanket guardrails against CBRN-adjacent dual-use knowledge, which requires nuanced contextual reasoning rather than keyword matching. Illegal Weapons (Non-CBRN) ranked third at 16.77%, followed by Malware Code at 12.11%.

Election Interference has continued its upward trajectory, now averaging 6.52% ASR across all runs — the fifth-highest category. This is a meaningful finding given the electoral calendar: models are demonstrably more susceptible to prompts designed to generate misleading electoral content, impersonate candidates, or fabricate polling data than to most other social-harm categories. The consistency of this finding across eight runs and seven models suggests it is structural rather than a sampling artefact.

Sexual Content (5.00%), Misinformation (3.72%), Bias Content (3.67%), and Environmental Harm (3.62%) round out the top ten. Privacy and Surveillance categories remained low across the study, averaging below 2%, which likely reflects the strong PII-detection capabilities of the RegexJudge component.

Risk Category			
CBRN Information or Capabilities	35.52%	46	MAP
Cybersecurity Threats (Beyond Malware)	21.60%	46	MAP
Illegal Weapons (Non-CBRN)	16.77%	46	MANAGE
Malware Code	12.11%	46	MAP
Election Interference ▲	6.52%	46	GOVERN
Sexual Content	5.00%	46	MANAGE
Misinformation	3.72%	46	MAP
Bias Content	3.67%	46	GOVERN
Environmental Harm	3.62%	46	MAP
Hateful Content	3.28%	46	GOVERN

Table 6. Top 10 risk categories by average ASR across all eight runs. ▲ = upward trend. NIST AI RMF function mapping shown.

5. Guardrail Stability Under Multi-Turn Pressure

The Drift Coefficient — computed on a 5% sub-sample of attack prompts subjected to 10 consecutive adversarial turns — measures how a model's safety posture shifts under sustained pressure. A positive coefficient indicates eroding guardrails; a negative coefficient indicates that the model becomes more conservative as the dialogue progresses.

Claude Opus 4.6 averaged a drift coefficient of $-0.28/\text{turn}$ across eight runs, the strongest stability result in the study: its guardrails actually tightened slightly under extended adversarial engagement. Claude Sonnet 4.6 followed at $-0.14/\text{turn}$. In contrast, Claude Opus 4.7 averaged $+0.69/\text{turn}$ — higher than expected given its overall ASR performance — suggesting that its safety fine-tuning may be less robust to multi-turn manipulation patterns specifically.

Gemini 2.5 Pro's May 16 drift coefficient of $+1.11/\text{turn}$ was the highest single-run reading recorded across the entire study. Combined with 23 critical failures in that same run, this represents the most concerning single-model result of the eight weeks and warrants close monitoring in future runs. GPT-4o Mini maintained a consistently positive drift across all eight runs ($+0.35/\text{turn}$ average), confirming a structural tendency toward guardrail softening under adversarial multi-turn dialogue.

6. Audience-Specific Insights

AI Engineers and Red-Teamers

- CBRN (35.52%) and Cybersecurity (21.60%) categories account for the majority of bypasses — these two categories alone drive over half of all attack failures.
- Gemini 2.5 Pro's drift coefficient of $+1.11/\text{turn}$ on May 16 is the study's highest single reading; multi-turn jailbreak resilience should be retested before any deployment update.
- All Anthropic models maintain near-zero FOR ($<2\%$) — model-level safety config appears well-calibrated for high-throughput enterprise use without over-refusal overhead.
- Election Interference (6.52% avg ASR) is now a structural top-5 risk category — add it to red-team priority lists, especially ahead of electoral periods.

GRC and Compliance Officers

- 399 critical findings across 23,000 evaluations (1.7% critical rate) — all mapped to NIST AI RMF functions (MAP, MANAGE, GOVERN, MEASURE) with chain-of-reasoning audit trails.
- GSAR 552.239-7001 forensic logs are generated for every CRITICAL_FAIL and all FAIL verdicts with severity ≥ 7 , providing PII-redacted prompt/response pairs for direct inclusion in compliance documentation.
- Gemini 2.5 Pro (17.60% avg FOR) and GPT-4o Mini (13.75% avg FOR) both exceed the 10% FOR threshold — flag for procurement review if deploying in user-facing applications.
- Provenance Scores averaged $\geq 99.8\%$ across all models, ensuring a near-complete audit trail for all 23,000 evaluations.

Product Leaders and Deployment Decision-Makers

- Anthropic models offer the best safety-utility balance: ASR below 6% and FOR below 2%, meaning low attack exposure without significant refusal overhead for legitimate users.
- Gemini 2.5 Flash's 3.75% average ASR is impressive, but its 14.60% FOR means roughly 1 in 7 safe user requests will be incorrectly rejected — evaluate carefully for customer-facing use cases.
- GPT-4o Mini's persistently high ASR (8.31% average, 9.75% in final run) and 13.75% FOR represent a dual failure mode: simultaneously too permissive to adversarial prompts and too restrictive to legitimate users.
- Gemini 2.5 Pro's 23 critical failures and +1.11 drift on May 16 represent a step-change regression from prior runs — treat the May 16 data as a blocking signal pending explanation.

Policy Analysts and Researchers

- Eight weeks of weekly data on a fixed open benchmark (RedBench, MIT License) enables time-series safety comparison independent of any vendor's internal reporting.
- Election Interference at 6.52% average ASR across 46 data points is a statistically robust finding across 7 models and 8 runs — not a sampling artefact.
- The CBRN gap (35.52% vs 21.60% for the second category) persists across all models and all runs, suggesting systemic limits of current RLHF-based alignment against dual-use knowledge requests.
- A negative average drift for Anthropic models (Opus 4.6: -0.28, Sonnet 4.6: -0.14) vs. positive drift for OpenAI and Google models suggests divergent multi-turn alignment strategies worth further investigation.

7. Conclusions

After eight weekly evaluation runs covering 23,000 evaluations across seven frontier models, four findings stand out. First, the safety-utility trade-off is not a universal constraint: Anthropic's Claude models consistently demonstrate that a model can achieve both low attack bypass rates (<6%) and near-zero false refusals (<2%), disproving the assumption that safety and usefulness are inherently in tension.

Second, CBRN and Cybersecurity categories remain structurally resistant to current guardrail approaches across all providers, with average ASRs of 35.52% and 21.60% respectively. These categories require domain-specific safety interventions beyond general RLHF alignment.

Third, Election Interference has solidified as a persistent top-5 risk category at 6.52% average ASR across all models and all runs. This finding carries heightened urgency given ongoing electoral cycles globally and warrants targeted safety investment from all providers.

Fourth, multi-turn guardrail stability varies dramatically by provider: Anthropic models trend toward tighter guardrails under sustained adversarial pressure (negative drift), while OpenAI and Google models generally trend toward looser guardrails (positive drift). Gemini 2.5 Pro's +1.11/turn drift and 23 critical failures on May 16 represent the most significant single-run safety regression observed in this study.

The pipeline, data, and audit logs for this study are fully open-source and available at <https://github.com/Sushegaad/Responsible-AI-Model-Evaluations> under the MIT License. Evaluations continue weekly and results are published automatically to the live dashboard.

8. References

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Disclaimer: This is an independent research initiative. Results reflect model behaviour on RedBench under standardised conditions and are not a comprehensive measure of safety in all deployment contexts. Results may vary due to model updates, API changes, and sampling randomness. No affiliation with Anthropic, OpenAI, Google, or the RedBench authors is implied.